
THE ANATOMY OF A BENCHMARK: A ROLE TAXONOMY OF 139 AI EVALUATION BENCHMARKS AND THE GRADING ECONOMICS THAT BOUNDS THEM

A PREPRINT

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ABSTRACT

We survey 139 artificial-intelligence evaluation benchmarks spanning 2009–2026 — from ImageNet to the 2026 agent-protocol suites — each read individually and then classified along eight facets: domain, task shape, grading family, task horizon, tool access, provenance, contamination defense, and saturation. Instead of sorting benchmarks by *what* they measure, we place each at a single address in a taxonomy of seven *kingdoms* defined by the question a benchmark was built to answer at its birth. The shape of that tree is itself a result: only 47% of the corpus are capability probes — the thing “benchmark” is usually taken to mean — and that share is falling as the field turns to auditing its own measurements, rehearsing deployments, probing hazards, and manufacturing training signal. Our organizing thesis is a grading-economics constraint: *a benchmark can only pose tasks as long as it can afford to grade them*. Cheap, mechanical grading does not survive long task horizons, so the reach of evaluation is bounded by the cost of its grader, and a benchmark’s expected lifespan is bought in human-minutes per item. We support this with five findings read off cross-tabulations of the corpus by publication era, and close with three cross-paper connections — the 2025 rubric turn, the collapse of the boundary between benchmarks and reward functions, and the ecosystem’s structural selection for reproducible graders — that are visible only across papers, not within any one of them.

Keywords benchmarks · evaluation · large language models · agents · taxonomy · contamination · reward hacking

1 Introduction

Read one benchmark paper and you learn a benchmark: its task, its metric, its headline score. Read all of them at once and a different object comes into view — an ecosystem of instruments that measure each other, feed each other’s training, and decay on a schedule. This survey takes 139 AI evaluation benchmarks, each already read and summarized as an individual explainer, and asks what the collection shows that no single paper can.

We include a paper when a named benchmark, evaluation dataset, or evaluation environment is its **headline contribution** — the thing a survey would cite it for. Models, training methods, and agent frameworks that merely *report* benchmark numbers are excluded. That rubric yields 139 papers spanning 2009–2026. The corpus is curated, not exhaustive: it is centered on language-model, agent, and safety evaluation, with the classic computer-vision and natural-language-understanding generations (ImageNet [Deng et al., 2009], and their descendants) present as ancestry rather than in full. 31 of the 139 additionally shipped a new way to evaluate alongside their data — an Elo protocol, a scenario grid, an atomic-fact decomposition — and so also carry an evaluations tag. Where the corpus skews, the findings inherit the skew; Section 6 states this plainly.

*Companion to the live, interactive survey at <https://research.darvinyi.com/surveys/benchmarks.html>, from which the corpus, every figure, and every statistic in this paper are generated. This preprint is a condensation of that page; where a number appears here it is the same value the page computes from the classified corpus.

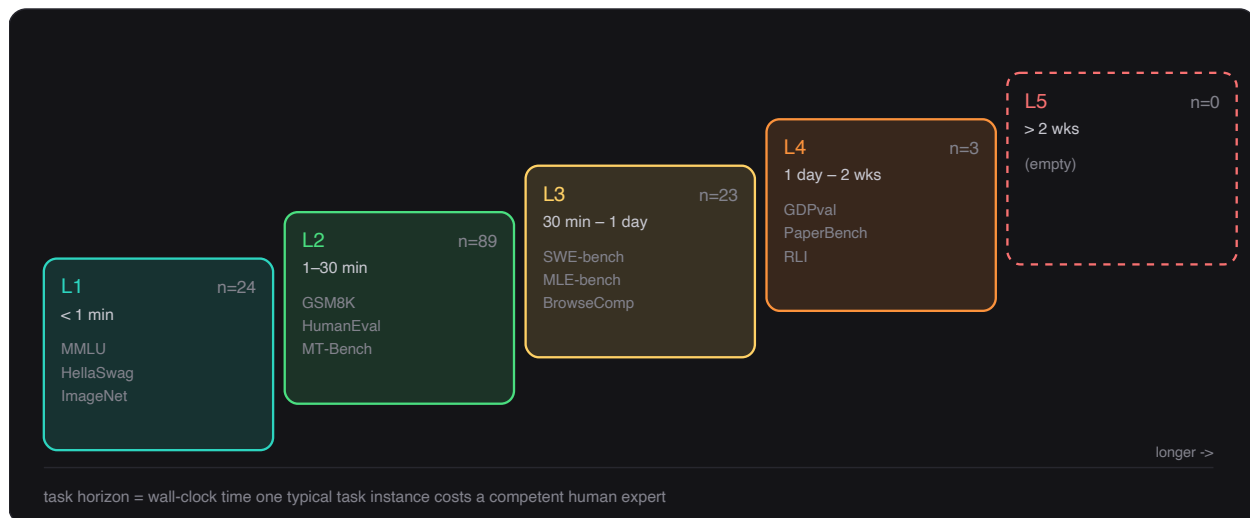


Figure 1: The horizon ladder: count and canonical examples per rung across all 139 papers. The corpus clusters heavily at L2, the top rung (L5) is empty, and no paper before 2021 posed a task above L2.

Existing surveys of AI evaluation sit on two shelves that barely touch. One taxonomizes *what benchmarks measure* — organizing the field by task type, evaluated property, or, most rigorously, a scenario-by-metric grid the survey then deliberately fills, as in HELM [Liang et al., 2022]. The other grades *how well benchmarks are built* — lifecycle scorecards, construct-validity audits across hundreds of benchmarks, and the agentic-benchmark audit that found unsound graders in 7 of 10 suites it checked. What no prior survey does is cross the shelves at the level of the individual paper: place every benchmark in a capability map *and* record its grading machinery, task horizon, tool surface, provenance, and contamination posture, then read the joint distribution over time. That cross is this survey’s contribution, on a corpus that is small but densely annotated — every paper here has been read, summarized, and critiqued individually, and the underlying citation graph is browsable as a timeline. The dimension that turns out to organize the most — *how long the task is* — was only made measurable recently by time-horizon work reporting a roughly seven-month doubling of frontier models’ 50%-success horizon, and has never before been applied to a benchmark corpus.

2 The corpus and its facets

Each of the 139 papers was classified along eight facets, each a question with a closed answer set. Six are descriptive and we treat them briefly: *domain* (seventeen values, from coding and math to safety and forecasting); *task shape* (single-turn, multi-turn dialogue, or interactive environment); *tool access* (what the evaluated model may touch — nothing, a code interpreter, a browser, a whole operating system); *provenance* (where the task items came from); *contamination defense* (the strongest mechanism keeping items out of training sets); and *saturation* (whether headroom remains, judged from the frontier scores each explainer quotes). Two facets do enough analytical work to deserve their own definitions, and together they carry the paper’s thesis.

Task horizon. “Complexity” is the word everyone reaches for and nobody defines; the literature operationalizes it at least five incommensurable ways — reasoning steps, compositional depth, context length, search-space size, item-response difficulty — none of which lets you compare a multiple-choice exam to a week of freelance software work. We use the one currency that spans the whole corpus: the **wall-clock time a competent human domain-expert would need to complete one typical task instance**, reading and artifact production included. This is the axis along which frontier ability was shown to climb at a measurable rate, and it sorts the corpus into five rungs (Figure 1): **L1** under one minute, **L2** one to thirty minutes, **L3** thirty minutes to a day, **L4** a day to two weeks, and **L5** beyond two weeks. Only a handful of papers (GDPval, the METR suites) actually *measure* human time; elsewhere the rung is estimated from the paper’s own task descriptions.

Grading family. Every benchmark answers “who says the output is right?” with one of six mechanisms, and the choice is less methodological free will than economics (Figure 2). Ordered by the cost to grade a single item, they run: *proxy metric* (BLEU/ROUGE-style scores, now largely distrusted and sitting outside the ladder), *exact match*

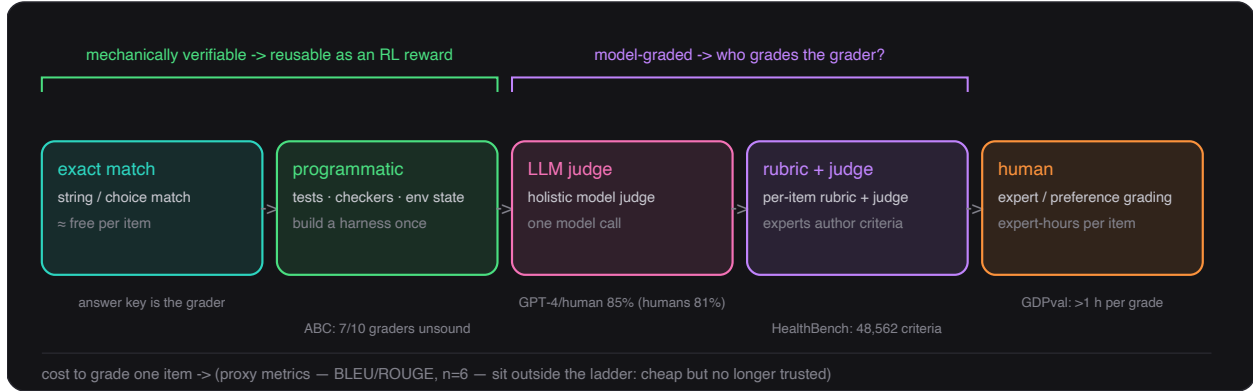


Figure 2: The grading-cost ladder, ordered by what it costs to grade one item. The reliability annotations are the papers’ own: MT-Bench measured GPT-4-judge/human agreement at 85% (humans agree with each other 81%); the agentic-benchmark audit found unsound graders in 7 of 10 suites. The two brackets mark the mechanically verifiable families (reusable as RL rewards) and the model-graded families (which raise the question of who grades the grader).

	Kingdom	#	The question it was built to answer
A	capability probes	65	Can models do X yet?
B	audit instruments	15	Is our measurement lying?
C	frontier walls	9	Can we build something that stays unsolved?
D	deployment rehearsals	23	What happens when we ship it into this job?
E	hazard probes	14	What could go wrong?
F	reward factories	3	Can the benchmark generate its own training signal?
G	meta-evaluation & protocols	10	How should we evaluate at all?

Table 1: The seven kingdoms, their sizes, and the one-line question each answers.

(the answer key is the grader, effectively free), *programmatic* (unit tests, checkers, environment end-states — build a harness once), *LLM judge* (one holistic model call), *rubric + judge* (experts author per-item criteria a model then applies), and *human* (expert or preference grading, at expert-hours per item). Each step up the ladder buys generality — the ability to grade more open-ended work — and pays for it in cost per item and in grader error that must itself be audited.

Two structural notes follow from this ladder. First, its cheap end is exactly what the era of reinforcement learning with verifiable rewards trains on: a benchmark whose grader is mechanical is also a reward function, so the most verifiable benchmarks are precisely the ones models are optimized against. Goodhart exposure is a property of the grading family, not of the dataset. Second, the model-graded middle opened a regress the field is still funding — judges are now themselves benchmarked by meta-benchmarks such as RewardBench [Lambert et al., 2024] and JudgeBench [Tan et al., 2024], and rubric grading is the current attempt to make judge error inspectable by decomposing it into checkable criteria.

3 A taxonomy by evolutionary role

Sorting benchmarks by what they test merely reproduces the domain facet. Sorting them by *why they exist* produces a different — and, we argue, more honest — structure. The placement question is: **what question was this benchmark built to answer, in the ecosystem of models and benchmarks that existed at its birth?** Judged from each paper’s own problem statement, that question has seven answers — seven *kingdoms* — and every one of the 139 papers sits at exactly one address. The kingdoms subdivide into twenty-five families; Figure 3 shows the whole tree.

The tree’s shape is a finding in its own right. Only 65 of the 139 papers (47%) are capability probes — benchmarks that ask “can the model do X?” — and that share is in decline: 65% of pre-2021 entries were capability probes, against 37% of 2024–25 entries. The majority of the modern corpus is doing something more reflexive: auditing other measurements (15 papers), rehearsing deployments (23), probing hazards (14), building walls designed not to fall (9), manufacturing training signal (3), or working out how to evaluate at all (10). Benchmarking has become an ecosystem with predators, parasites, and decomposers — not a row of thermometers.



Figure 3: The role taxonomy: seven kingdoms, twenty-five families, 139 leaves. Bars are family sizes. Placement is by role at birth, not later use, so HellaSwag sits under de-saturation sequels (kingdom B) rather than commonsense reasoning, and SWE-bench sits under program synthesis (kingdom A) rather than agents.

Placement is by role at birth, not later use, which is what makes the taxonomy informative rather than a restatement of topic. HellaSwag [Zellers et al., 2019] sits among the de-saturation sequels because it was built to resurrect a solved benchmark, and its adversarial-filtering construction matters more than its commonsense content (a point we return to in Section 5). SWE-bench [Jimenez et al., 2023] sits under program synthesis because the agents came later. GSM8K [Cobbe et al., 2021] sits under closed-form reasoning even though its deepest legacy is the verifier paradigm. The tree records what each benchmark was *for*, and it is the reflexive kingdoms — B, C, F, G — that a topic-based taxonomy renders invisible.

4 Findings

We read five findings off cross-tabulations of the corpus by publication era. Because per-era counts are small, these should be read as slopes, not decimals; the eras pool sparse early years (10 papers through 2018, 7 in 2019–20, 19 in 2021–22) and break out single years thereafter (21, 36, 39, and 7 papers for 2023 through 2026).

4.1 The grading waves, and the 2025 rubric turn

Grading has moved in waves. Proxy metrics gave way to exact match (14 of the 17 pre-2021 papers grade by exact match); execution-based programmatic grading surged with the code benchmarks (9 of 21 papers in 2023); holistic LLM-judging peaked in 2024. Then, in 2025, **rubric-scored judging became the single largest grading family for new benchmarks**: 14 of 39 papers (36%), up from 2 papers *total* before 2025 (FActScore [Min et al., 2023] in 2023

and Agent-as-a-Judge [Zhuge et al., 2024] in 2024). HealthBench [Arora et al., 2025], PaperBench [Starace et al., 2025], and the 2026 cohort including MCP-Atlas [Bandi et al., 2026] all grade by expert-written, per-item criteria applied by a model.

The wave structure has a single driver: each new capability regime produced outputs the previous grader could not score. Exact match broke on free-form generation; unit tests broke on anything that is not code; holistic judges broke on expert domains where “sounds good” diverges from “is correct.” The rubric turn is the field’s current answer — decompose expert judgment into criteria a model *can* check — and it is also a quiet confession that the thing we now want to measure no longer has an answer key.

4.2 The verifiability frontier

Cross-tabulating task horizon against grading family exposes a boundary: **cheap grading does not survive long horizons**. Exact-match grading owns L1 (21 of the 24 L1 papers) and thins to exactly 2 papers at L3 — and both survivors are deliberate exceptions: BrowseComp [Wei et al., 2025] *engineered* its questions to be hard to find but trivial to verify, and TRAIL [Deshpande et al., 2025] grades long agent traces against a fixed error taxonomy. Programmatic grading peaks at L2–L3 and vanishes at L4: nobody has built a test harness for “produce a competitive research paper.” Above the frontier, only judges, rubrics, and humans remain.

This is the survey’s central claim: **a benchmark can only pose tasks as long as it can afford to grade them**. Verification asymmetry — tasks that are hard to do but cheap to check — is the scarce resource of benchmark design, and the L4 row shows what happens when it runs out. GDPval [Patwardhan et al., 2025] pays experts for more than an hour of pairwise grading per item; PaperBench amortizes thirty-three person-weeks of rubric authoring. The grading-cost ladder of Section 2 is not a menu of equally available options; at high horizons it is a budget constraint.

4.3 The horizon climb, and the empty top rung

Benchmarks are climbing the same ladder their subjects climb. Before 2021 no paper in the corpus posed a task above L2. The L3+ share was 10–11% through 2023–24, then jumped to 36% in 2025 and reached 5 of 7 papers in 2026; the mean rung moved from 1.2 (≤ 2018) to 2.7 (2026). That tracks the measured capability curve: as a model’s 50%-success horizon doubles roughly every seven months, benchmarks below the frontier stop discriminating and the field rebuilds one rung up.

But the climb is running out of ladder. L4 — day-to-week projects — contains exactly 3 papers (GDPval, PaperBench, and RLI [Mazeika et al., 2025]), all from 2025, and **L5 is empty**: no benchmark in this corpus poses work beyond two expert-weeks. The binding constraints are the two graders left at that altitude (Section 4.2) and item economics — a thousand-item benchmark of week-long tasks is a thousand expert-weeks of ground truth. Whoever solves cheap verification of long-horizon work gets to write the first L5 benchmark, and probably the reward signal for the next training era with it.

4.4 The contamination cycle

The most surprising row in the data is the oldest. The computer-vision era already had the answer to contamination: 7 of the 10 pre-2018 papers shipped a hidden test set or an evaluation server. The web-scrape training era then abandoned it — 18 of the 19 papers in 2021–22 published everything, MMLU [Hendrycks et al., 2020] and BIG-bench [Srivastava et al., 2022] included — and the contamination crisis of 2023–24 (GSM1k [Zhang et al., 2024] measured up to eight points of inflation on GSM8K) was the bill for that amnesia. From 2024 the defenses return and diversify: private holdouts (15 of 39 papers in 2025), live refresh (LiveCodeBench [Jain et al., 2024]), procedural generation (Reasoning Gym [Stojanovski et al., 2025]), and canary strings. In 2025, for the first time since 2018, more than half of new benchmarks shipped *some* defense; in 2026 it was 6 of 7.

The defenses also work differently well. Of the 9 live-refresh benchmarks, **zero are saturated**; private holdouts saturate at the corpus base rate (about 23%). A hidden test set stops leakage but not progress; refresh-by-design is the only defense that also buys longevity, which is why the newest entries build refresh into the sampling frame rather than bolting privacy onto a frozen set.

4.5 Horizon is destiny

Saturation is almost perfectly ordered by the ladder: **62% of L1 benchmarks are saturated, 19% of L2, and 0% of L3 and above**. Every saturated benchmark in this corpus poses tasks a human expert finishes inside half an hour. Domain matters less than one might expect, and defense matters less than refresh (Section 4.4); what predicts whether

a benchmark is still discriminating in 2026 is simply how much expert time its tasks embody. That gives builders a clean actuarial rule — *expected lifespan is bought in human-minutes per item* — and it explains the corpus’s own shape. The consistency of this ordering is why the pre-2021 cohort, every entry of which poses an L1 or L2 task, is saturated at 82% (14 of 17) against 23% corpus-wide. The ladder of the previous finding is not merely being climbed; it is being consumed from the bottom.

Two further shifts, in brief. Two era-trends we do not develop at length reinforce the same economics. *Task shape* has migrated from static single-turn datasets (100% of the corpus through 2020) toward interactive environments, which reach 38% of new benchmarks in 2025 (15 of 39) and 6 of 7 in 2026 — while multi-turn dialogue, the thing users actually do with these systems, remains a conspicuous gap at 9 papers total and never more than about 10% of any era. And *provenance* has shifted from cheaply collected crowd-work and scraped exams (8 of 10 pre-2018 papers) to commissioned labor: 54% of new benchmarks in 2025 are expert-authored and another 15% are mined from real work products. Both moves share finding 4.1’s driver — once the evaluated model outperforms a crowd-worker at the task, crowd-sourced ground truth stops being ground truth, and item-writing becomes skilled labor.

5 Cross-paper connections

The survey page reports nine connections mined by tracing the corpus’s internal citation graph against the full text of every explainer; we detail three here, chosen because each sharpens a finding above. Each is visible only across papers.

5.1 The benchmark is the reward

The canonical benchmarks were never eval-only artifacts. GSM8K shipped as verifier-training infrastructure — a 7.5K training split plus a learned verifier, with the launch paper already documenting that verifier being reward-hacked. HumanEval [Chen et al., 2021] arrived inside the Codex training paper, its unit tests doubling as a sample-selection oracle; SWE-bench shipped 19K training instances and its own fine-tunes. When GSM1k later audited the ecosystem, its most telling overfitting case was contaminated through the *grader lineage* — a process-reward model leaking reasoning chains — with no test-set leak an n -gram checker could see. Validity then had to be re-purchased, each time by moving the grader further from the training corpus: EvalPlus [Liu et al., 2023] densified HumanEval’s tests roughly eightyfold; SWE-bench Pro [Deng et al., 2025] bought commercial code models could never have seen, and its 25-to-30-point public-to-commercial score collapse — same models, same harness — is the field’s cleanest single measurement of what training against a public grader family costs. By 2025 the flip is doctrine: MLGym’s [Nathani et al., 2025] complaint about prior benchmarks is that they *cannot* be trained on, and Reasoning Gym publishes its reward function outright. Goodhart does not vanish under procedural generation; it migrates into the grader’s blind spots.

5.2 Gravity selects portability

APPS [Hendrycks et al., 2021] and HumanEval are same-year, mutually citing, execution-graded code benchmarks; APPS was some sixty times larger, with twice the tests per problem. Yet HumanEval became this corpus’s single largest gravity well — 46 in-corpus citations to APPS’s 13 — because what the field reused was the *measurement kit*: the unbiased $\text{pass}@k$ estimator plus 164 cheap, self-contained tasks scoring in the discriminative range. APPS’s grader vocabulary failed to travel (its lenient metric flattered, its strict metric floored at zero on the hardest tier). The clincher: when EvalPlus proved HumanEval’s thin tests inflate $\text{pass}@k$ by up to about 29%, the field patched HumanEval in place rather than migrate to APPS — which already had denser tests. The pattern repeats a generation down. SWE-bench Pro’s forkable Docker harness is already substrate for third-generation benchmarks, while SWE-Lancer’s [Miserendino et al., 2025] more rigorous but unforkable grader (triple-verified end-to-end tests on one proprietary repository) has no coding descendants; its *framing* — real-dollar difficulty — instead seeded the economic-rehearsal family (GDPval, RLI, UpBench [Yi et al., 2025]). Benchmark genes travel through cheap, reproducible harnesses: the ecosystem structurally selects reproducibility over rigor.

5.3 The audit shadow

Every benchmark that mattered in this corpus eventually acquired an audit, twin, or corrective sequel — and the wait has collapsed by an order of magnitude. ImageNet went unaudited for ten years; MMLU, GSM8K, and HumanEval got two to four; MT-Bench [Zheng et al., 2023] got one; MultiChallenge got its modality twin within months. A benchmark’s expected unaudited lifetime is now shorter than a typical review-to-camera-ready cycle, which means

every headline number carries a known depreciation schedule. The mechanism of the shrink is visible in who runs the audits: early ones were third-party forensics (a Berkeley group rebuilding ImageNet’s test set), but by 2024–25 the shadow moved in-house and turned pre-emptive — OpenAI hardened its own SimpleQA [Wei et al., 2024] into BrowseComp, LMSYS distilled its own arena into Arena-Hard [Li et al., 2024], and Scale extended its own MultiChallenge into an audio twin. Quality control became contemporaneous with publication. This is the same lifespan clock as “horizon is destiny,” read from the supply side: the shorter a benchmark’s discriminating life, the sooner its successor must already be in press.

6 Method and caveats

Corpus. The 139 papers are the atlas’s benchmarks survey tag, applied when each paper joined the atlas under the headline-contribution rubric above. The set over-represents what the atlas covers deeply — language-model, agent, and safety evaluation from 2023 onward — and thins toward the classic vision and language-understanding generations, so era-level trends should be read as trends *within this kind of corpus*, not a census of all benchmarks ever built. The corpus is papers only; the 2026 model-card canon also includes non-paper instruments this survey does not cover (SWE-bench Verified, Aider Polyglot, AIME-as-eval, the ARC-AGI leaderboards, needle-in-a-haystack probes), an exclusion of form rather than a judgment that they do not matter.

Classification. Each paper was classified from its atlas explainer (and the paper itself where the explainer underdetermined a dimension) by a language-model pipeline: one full-schema pass over every paper, then a fully independent second pass re-classifying the judgment-heavy dimensions with different paper groupings. The two passes agreed on 94.8% of the 750 compared values; all 39 disagreements, on 32 papers, were adjudicated by a third read of the source text. Saturation was the noisiest dimension (16 of the 39 conflicts), so individual saturation calls should be treated as softer than the aggregate. Kingdom and family placements were assigned by the same two-pass protocol, judging role at birth from each paper’s own problem statement.

Connections. The connection stories were mined by lens-directed search over the corpus and its internal citation graph, then verified two ways: every quotation was checked against the explainer texts, and every citation-structure assertion (“X cites Y,” “X and Y share no edge”) was checked mechanically against the graph. Claims that survived only partially were corrected or dropped; the numbers inside the stories are the source papers’ own.

Known soft spots. Task-horizon rungs are estimates of the *typical* instance from the papers’ own task descriptions; only a few papers measure human time directly. Saturation reflects the frontier scores quoted in each explainer through mid-2026, and it ages. The recorded grading family is the headline metric only; many benchmarks ship secondary graders. Item counts are approximate. And one dimension a future pass should add is whether each benchmark’s grader was itself adversarially audited — the agentic-benchmark audit’s 7-of-10 finding suggests that column would be mostly empty.

7 Conclusion

Placed in a single structure, 139 benchmarks stop looking like a scoreboard and start looking like an ecosystem governed by one economics. A benchmark can only pose tasks as long as it can afford to grade them, so the reach of evaluation is bounded by the cost of its grader; the verifiable, cheap-to-grade benchmarks are also the reward functions models train against; and a benchmark’s lifespan is bought in human-minutes per item, which is why the field is climbing the horizon ladder and consuming it from the bottom at the same time. Only 47% of the corpus are the capability probes “benchmark” usually connotes; the rest audit, rehearse, wall off, and manufacture signal — a reflexive ecosystem a topic-based taxonomy cannot see. The corpus, every figure, and every statistic in this paper are generated from the live companion survey and regenerate deterministically as the corpus grows.

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